Problem Set for 10 April 2025

Notation Throughout, we denote by $\hookrightarrow$ a linear injection, $\twoheadrightarrow$ a linear surjection, and $\simeq$ or $\xrightarrow{\sim}$ a linear isomorphism. For simplicity, we write $(U, V) =: \mathrm{Hom}_{\mathbb{F}}(U, V)$.

Exercise 1 Translation.

S	Functorial Language	Natural Language
1	$(\varphi \circ -) : (U, V_1) \twoheadrightarrow (U, V_2)$	Every $U \xrightarrow{f} V_2$ factors through $V_1 \xrightarrow{\varphi} V_2$.
2	$(\varphi \circ -) : (U, V_1) \hookrightarrow (U, V_2)$	?
3	$(- \circ \psi) : (U_1, V) \twoheadrightarrow (U_2, V)$	?
4	$(- \circ \psi) : (U_1, V) \hookrightarrow (U_2, V)$	Any $U_2 \xrightarrow{f} V$, which factors through $U_2 \xrightarrow{\psi} U_1$, factors through ψ uniquely.
5	$\ker \left[(U, V_1) \xrightarrow{\varphi \circ -} (U, V_2) \right]$	?
6	$\mathrm{im} \left[(U, V_1) \xrightarrow{\varphi \circ -} (U, V_2) \right]$	The subspace of (U, V_2) consisting of the linear maps which factor through $V_1 \xrightarrow{\varphi} V_2$.
7	$\ker \left[(U_1, V) \xrightarrow{- \circ \psi} (U_2, V) \right]$	The subspace of (U_1, V) consisting of the linear maps whose kernel contains the image of ψ.
8	$\mathrm{im} \left[(U_1, V) \xrightarrow{- \circ \psi} (U_2, V) \right]$	?

It is highly recommended to explain 3E-19 (linear algebra done right) in functorial language.

Exercise 2 Suppose that $K \subseteq U$ is the kernel of $U \xrightarrow{f} V$. Denote the inclusion by the linear map $i : K \hookrightarrow U$. Show that for any linear space W, the linear map

$$(f \circ -) : (W, U) \to (W, V), \quad \varphi \mapsto f \circ \varphi$$

has kernel (W, K) along with the inclusion

$$(i \circ -) : (W, K) \hookrightarrow (W, U), \quad \varphi \mapsto i \circ \varphi.$$

Show in steps that

Step 1 $(i \circ -)$ is indeed an injection, identifying (W, K) as a subspace of (W, U).

Step 2 Show that $\ker(f \circ -)$ coincides with the image of $(i \circ -)$, that is, the subspace (W, K).

Exercise 3 Suppose that $V \twoheadrightarrow \frac{V}{\operatorname{im} f}$ is the quotient map induced by a given $U \xrightarrow{f} V$. Denote the natural quotient map by the linear map $\pi : V \twoheadrightarrow \frac{V}{\operatorname{im} f}$. Show that for any linear space X, the linear map

$$(- \circ f) : (V, X) \to (U, X), \quad \psi \mapsto \psi \circ f$$

has kernel $(V/(\operatorname{im} f), X)$ along with the inclusion

$$(- \circ \pi) : (V/(\operatorname{im} f), X) \hookrightarrow (V, X), \quad \psi \mapsto \psi \circ \pi.$$

Show in steps that

Step 1 $(- \circ \pi)$ is indeed a surjection, identifying $(V/(\operatorname{im} f), X)$ as a subspace of (V, X).

Step 2 Show that $\ker(- \circ f)$ coincides with the image of $(- \circ \pi)$, that is, the subspace $(V/(\operatorname{im} f), X)$.

Exercise 4 According to **Exercises 2-3**, translate the following statements into mathematical expressions (functorial language as mentioned in **Exercise 1**):

- For any linear map f, the linear maps annihilated by the pre-composition $(f \circ -)$ identify the linear maps with codomain $\ker f$.
- For any linear map f, the linear maps annihilated by the composition $(- \circ f)$ identify the linear maps with domain $\frac{\operatorname{codomain} f}{\operatorname{im} f}$.

Fact (optional) Given $f : X \to Y$, we define

1. $\ker f$ (kernel) the natural inclusion map $\{x \in X \mid f(x) = 0\} \hookrightarrow X$, or just the subset $\{x \in X \mid f(x) = 0\}$ for simplicity;
2. $\operatorname{cok} f$ (cokernel) the natural quotient map $Y \twoheadrightarrow \{y + \operatorname{im} f \mid y \in Y\}$, or just the quotient set $\{y + \operatorname{im} f \mid y \in Y\}$ for simplicity.

For any Z, one has the natural isomorphisms

$$(Z, \ker f) \simeq \ker \left[(Z, X) \xrightarrow{(f \circ -)} (Z, Y) \right], \quad \left[Z \overset{\varphi}{\to} \ker f \right] \mapsto \left[Z \overset{\varphi}{\to} \ker f \hookrightarrow X \right],$$

and

$$(\operatorname{cok} f, Z) \simeq \ker \left[(Y, Z) \xrightarrow{(- \circ f)} (X, Z) \right], \quad \left[\operatorname{cok} f \overset{\psi}{\to} Z \right] \mapsto \left[Y \twoheadrightarrow \operatorname{cok} f \overset{\psi}{\to} Z \right].$$

The kernel objects and cokernel objects are defined not only for linear spaces but also for the entirety of mathematical structures, serving as solutions to the *problème d'application universelle*. For instance, if one seeks a subset of linear maps (X, Y) that send a given subset $S \subseteq X$ to zero, then $(X/\operatorname{span}(S), Y)$ precisely fulfils this requirement (by pre-composing the natural quotient map $X \twoheadrightarrow X/\operatorname{span}(S)$).

Analogously, the collection of Hom-spaces $(X, Y_\lambda)_{\lambda \in I} = \prod_{\lambda \in I}(X, Y_\lambda)$ corresponds to a single Hom-space $(X, \prod_{\lambda \in I} Y_\lambda)$, whereas the collection $(X_\lambda, Y)_{\lambda \in I} = \prod_{\lambda \in I}(X_\lambda, Y)$ corresponds to a single Hom-space $(\coprod_{\lambda \in I} X_\lambda, Y)$ due to logical considerations.